

Lecture 5

September 8

Probably...

What is probability?

— the chance of something happening
an "event" happening

— A number between 0 and 1 describes the } non-negativity
prob. of something.

— A coin toss: Heads vs Tails

$$Pr(\text{Heads}) = 0.5$$

$$Pr(\text{Tails}) = 0.5$$

$$\left. \begin{array}{l} \text{total prob} \\ = 1 \end{array} \right\} \underline{\quad}$$

Additivity: $Pr(\text{Heads or Tails}) =$
 $Pr(\text{Heads}) + Pr(\text{Tails})$

Axioms in blue

Learning objectives

- Define probability and distinguish forward (a priori) probability questions from backward (a posteriori) inference.
- Define Bernoulli and binomial random variables, including their probability mass functions and expectations.
- Explain what the law of large numbers says about relative frequencies and uncertainty—and what it does not say about individual outcomes or finite samples.

~~A priori~~ vs ~~a posteriori~~
Forward Backward/inference

- F: Given or known to contain r red balls and s black balls, the chance of a red ball being drawn is $r/r+s$

Conditions of a model $\rightarrow$ data

- B: Determine r and s based on observations of the game.

data $\rightarrow$ conditions/parameters of a model.

What is the probability of getting Heads
when flipping a coin?
50/50

Coin-tossing is a basic example of a random phenomenon. However, naturally tossed coins obey the laws of mechanics (we neglect air resistance) and their flight is determined by their initial conditions. Figure 1 a-d shows a coin-tossing machine. The coin is placed on a spring, the spring released by a ratchet, the coin flips up doing a natural spin and lands in the cup. With careful adjustment, the coin started heads up always lands heads up – one hundred percent of the time.

Dynamical bias in the coin toss
Diaconis, Holmes, Montgomery

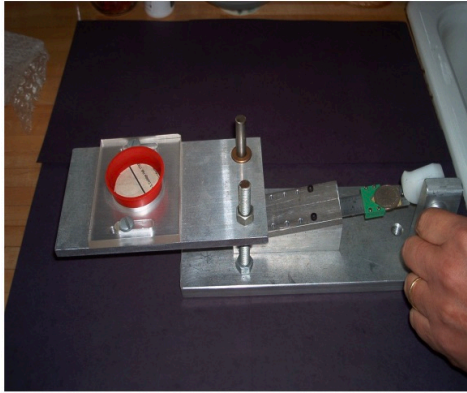


Figure 1.a

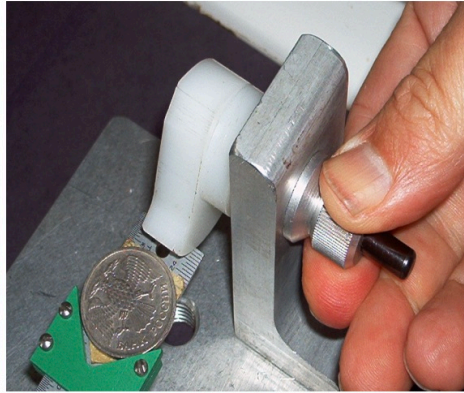


Figure 1.b

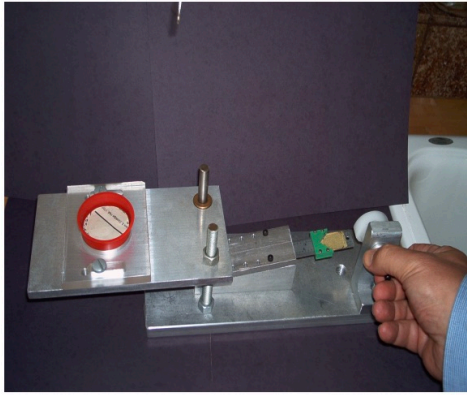


Figure 1.c

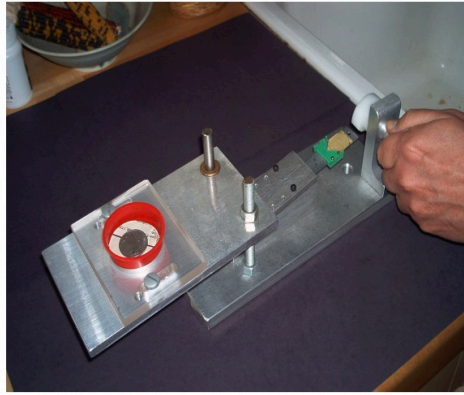
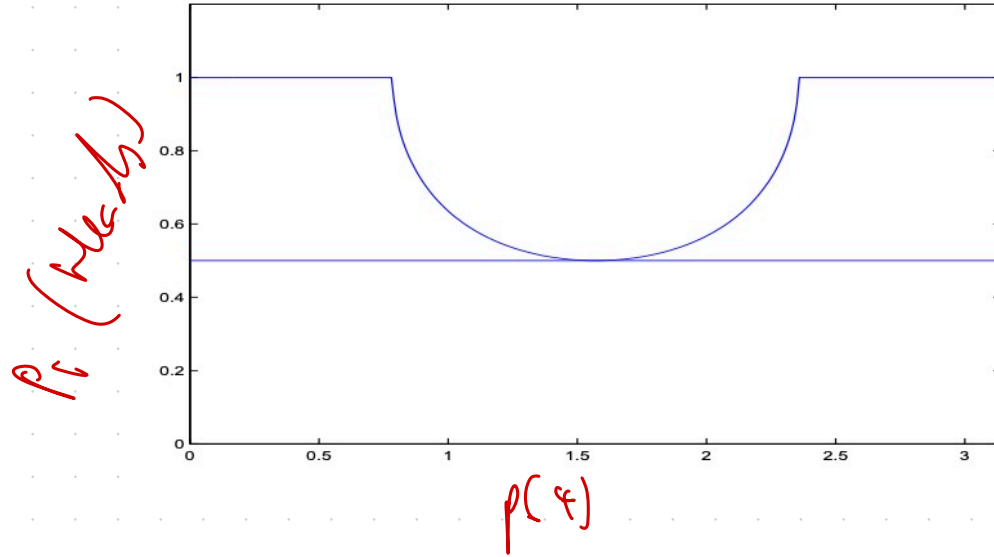


Figure 1.d

Coin toss machine

Start with heads
End with heads



The empirical estimates of ψ show that naturally flipped coins precess sufficiently to force a bias of at least .01. We find it surprising that this bias persists in the limit of vigorously flipped coins for general densities $g(\omega_N, t)$.

Our first distributions: the Bernoulli and the Binomial
Coin is flipped 10 times and lands heads 7 times.

Outcomes: H / T

Count: 7 occurrences

Proportion: $7/10$

Let Y be a Bernoulli random variable

$$\Rightarrow Y = \begin{cases} 1 & \text{with prob } p \in (0,1) \\ 0 & \text{with prob } 1-p \in (0,1) \end{cases}$$

$$Pr(Y=y) = p^y (1-p)^{1-y} \quad \text{prob. mass function}$$

Multiple observations of a coin

Let $Y_1, Y_2, Y_3, \dots, Y_n$ be independent Bernoulli (p)

$X = Y_1 + Y_2 + \dots + Y_n = \sum_{i=1}^n Y_i$ random variable

$$P(X = k) = ?$$

	0	1
0	$(1-p)^2$	$p(1-p)$
1	$(1-p)p$	p^2

$$P(Y_1 = 0, Y_2 = 0) = (1-p) \cdot (1-p)$$

by independence $= P(Y_1 = 0) P(Y_2 = 0)$

$$P(Y_1 = 0, \dots, Y_n = 0) = P(Y_1 = 0) \cdot \dots \cdot P(Y_n = 0)$$

of ways
to get
 k 1's out
 $n-k$ 0's.

$$P(X = k) = \underbrace{\quad}_{\text{independence}} p^k (1-p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Properties of distributions

For a Binomial (n, p) — n independent trials
each with prob p of success.

What is the expected # of successes.

Expectation of a random variable

$$E[X] = n \cdot p$$

$$E\left[\frac{X}{n}\right] = \frac{n \cdot p}{n} = p$$

What is "identifiability"?

Jacob Bernoulli

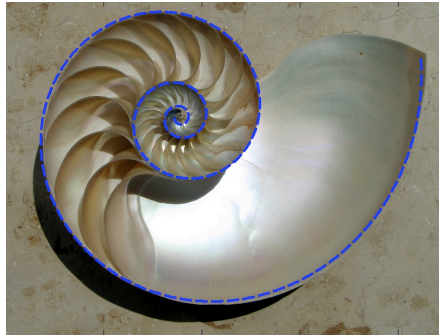
1654/5 – 1705, Basel, Switzerland

Destined for the ministry...

In vivo patre sidera verso

Brotherly love?

In life and death...



Goal: determine the probability of a fertile case
a posteriori.

Establish certainty about how many observations
are needed to study this probability.

What is moral certainty?

— certainty such that you are
willing to take actions in real life

⋮
but with math

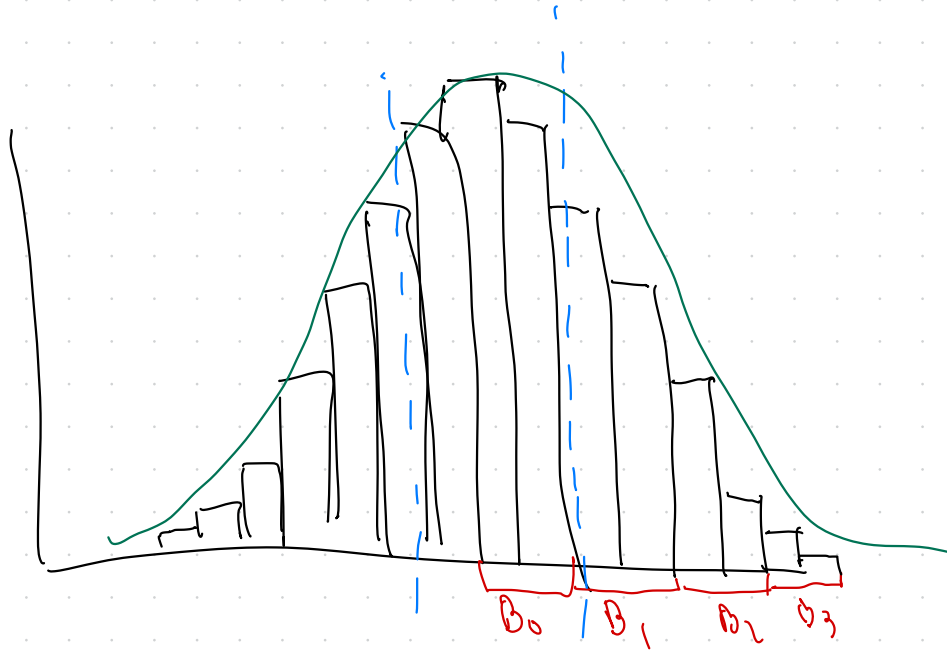
To illustrate this by an example, I suppose that without your knowledge there are concealed in an urn 3000 white pebbles and 2000 black pebbles, and in trying to determine the numbers of these pebbles you take out one pebble after another (each time replacing the pebble you have drawn before choosing the next, in order not to decrease the number of pebbles in the urn), and that you observe how often a white and how often a black pebble is withdrawn. The question is, can you do this so often that it becomes ten times, one hundred times, one thousand times, etc., more probable (that is, it be morally certain) that the numbers of whites and blacks chosen are in the same 3 : 2 ratio as the pebbles in the urn, rather than in any other different ratio? (Bernoulli, 1713, pp. 225–226)

Law of Large Numbers

$$P(|\bar{X}/n - p| \leq \varepsilon) > c P(|\bar{X}/n - p| > \varepsilon)$$

Usually $P(|\frac{\bar{X}}{n} - p| > \varepsilon) \xrightarrow[\text{converges}]{} 0$ as $n \rightarrow \infty$ for all $\varepsilon > 0$

Proof illustration



Why was Bernoulli unhappy?

Guarantee that having $N = 25550$ observations
means we can say that p is within $\frac{1}{50}$ of $\frac{30}{50}$
with probability $\frac{1000}{1001}$

De Moivre

Slightly different problem:

Given N , find the probability.

Next
claim